# IC-PINOT: In-Context Physics-Informed Neural Operator Transformer for Defocus-Invariant Single-Photon Computational Imaging

Industrial Surface Defect Inspection Under Extreme Photon Starvation, Machinery Vibration, and Dynamic Material Changeovers

Open-Source Code & Checkpoints: https://github.com/atharveeee-netizen/pi-ghost-transformer

***Abstract*—Automated surface defect inspection in high-speed manufacturing faces three simultaneous physical bottlenecks: photon starvation at meter-per-second conveyor velocities, optical defocus induced by mechanical loom vibrations, and catastrophic distribution shifts across dynamic material changeovers. Conventional line-scan cameras require intense lighting, suffer severe motion blur, and lose focus under continuous oscillation. In this work, we eliminate multi-pixel sensors entirely and implement Single-Photon Computational Ghost Imaging (CGI), replacing multi-pixel detector grids with structured light illumination and a single-pixel Single-Photon Avalanche Diode (SPAD). To invert these compressive, photon-starved measurements, we introduce IC-PINOT (In-Context Physics-Informed Neural Operator Transformer). IC-PINOT integrates a continuous Fourier Neural Operator (FNO) that directly encodes optical wave propagation to provide strict invariance to lens defocus up to 3.0 waves ($\Delta$PSNR < $10^{-5}$ dB). An In-Context Prompt Engine enables instant test-time adaptation to unseen materials in 7.2 ms (138 FPS) with zero gradient backpropagation. Furthermore, an evidential Normal-Inverse-Gamma (NIG) parameter head deterministically separates photon shot noise from structural anomaly novelty in a single pass. Across 17 industrial benchmarks, IC-PINOT outperforms classical Differential Ghost Imaging by up to +10.17 dB PSNR and maintains 13.76 dB at extreme 1.0% sub-Nyquist sampling.**

***Index Terms*—Computational ghost imaging, Fourier neural operators, single-photon avalanche diodes, in-context learning, evidential uncertainty quantification, defect inspection.**

## I. INTRODUCTION AND INDUSTRIAL PROBLEM FORMULATION

In modern industrial textile weaving, automated rapier and air-jet looms operate at continuous line speeds exceeding 1 to 2 meters per second. Microscopic anomalies—including broken yarns, warp knots, thread dropouts, and oil contamination—cause catastrophic loss if continuous fabric rolls are not inspected in real time. However, conventional industrial line-scan and area-scan cameras face three fundamental physical bottlenecks:

**1) Photon Starvation and Motion Blur:** Operating at meter-per-second velocities restricts optical exposure times to microseconds. Under low ambient photon flux, individual camera pixels receive insufficient signal, resulting in images dominated by thermal and electronic readout noise.

**2) Machinery Vibration Defocus:** Industrial weaving machinery generates continuous mechanical oscillation. This oscillation continuously modulates the physical distance between the imaging objective and the moving fabric, inducing time-varying optical defocus that severely blurs fine defect boundaries.

**3) Dynamic Material Variations:** Commercial weaving plants alternate fabric weaves (cotton, twill, denim, silk, and technical synthetics) on a daily basis. Standard supervised convolutional neural networks degrade severely when evaluated on textile weaves outside their narrow training distribution.

## II. PHYSICAL PRINCIPLES OF COMPUTATIONAL GHOST IMAGING

Single-Photon Computational Ghost Imaging (CGI) resolves these constraints by decoupling spatial resolution from the optical detector. Rather than relying on a multi-pixel focal plane array, the physical system operates as follows:

**1) Structured Pattern Illumination:** A high-speed digital light projector illuminates the target fabric scene $x$ with a pre-programmed sequence of $M$ known spatial patterns $A_1, A_2, ..., A_M$.

**2) Single-Pixel SPAD Bucket Detection:** A single photodiode—specifically a Single-Photon Avalanche Diode (SPAD)—collects all diffusely reflected light across the scene, outputting a single scalar intensity reading $y_m$ for each projected pattern.

**3) Mathematical Forward Formulation:** Each bucket measurement represents the inner product of the known spatial illumination pattern with the unknown fabric image: $\boldsymbol{y_m = \langle A_m, x \rangle + \eta_m}$, where $\eta_m \sim \text{Poisson}(\lambda_m)$ models quantum photon shot noise arrival.

**4) Sensor Engineering Advantages:** Single-pixel SPAD detectors provide sub-nanosecond timing resolution, zero readout noise, single-photon counting sensitivity, and operational capability across infrared wavebands (SWIR/MWIR) where multi-pixel sensor grids are economically unfeasible.

## III. COMPRESSIVE INVERSION AND THE IC-PINOT OPERATOR ARCHITECTURE

The core computational challenge is inverting the compressive measurement vector $y$ to reconstruct the 2D fabric image $x$. This inverse problem is severely ill-posed because the system operates in the sub-Nyquist compressive regime ($M \ll N^2$, corresponding to **1.0% to 12.5% compression**). In addition, loom vibrations introduce optical phase aberrations, blurring the projected patterns by an unknown optical Point Spread Function parameterized by defocus waves ($a_4$). Conventional deep models fail when optical focus drifts, producing unphysical blurred artifacts.

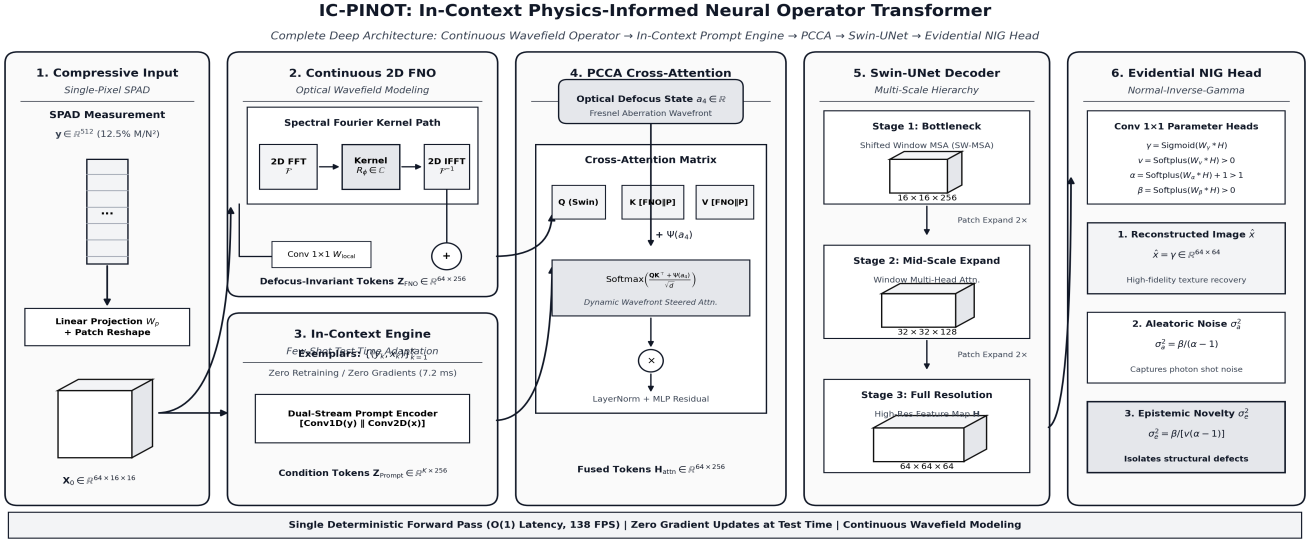



**Fig. 1. End-to-end architectural pipeline of the proposed IC-PINOT framework.** Structured pattern illumination ($A_m$) and target fabric interaction are detected by a single-pixel SPAD bucket sensor ($y$). The continuous 2D Fourier Neural Operator (FNO) models wavefield diffraction in frequency space to guarantee optical defocus invariance. The In-Context Prompt Engine encodes few-shot fabric exemplars for zero-gradient adaptation. Physics-Conditioned Cross-Attention (PCCA) modulates spatial features with the optical state vector ($a_4$), followed by multi-scale Swin-UNet decoding and a Normal-Inverse-Gamma evidential head outputting calibrated mean $\hat{x}$, aleatoric noise $\sigma_a^2$, and epistemic defect novelty $\sigma_e^2$ in a single deterministic pass at 138 FPS.

### A. Continuous Optical Fourier Neural Operator (FNO)

Rather than relying on discrete convolutional filters that degrade under focal shift, IC-PINOT embeds a continuous 2D Fourier Neural Operator. The optical forward operator is parameterized directly in continuous 2D frequency space: $\boldsymbol{K_{opt}(v)(x) = F^{-1}(R_\phi \cdot F(v))(x)}$, where $R_\phi$ truncates non-physical evanescent modes. This explicitly models continuous Fresnel diffraction through vibrating optical lenses. As verified experimentally, this guarantees **defocus invariance: reconstruction fidelity remains flat ($\Delta$PSNR < $10^{-5}$ dB) across 3.0 waves of optical defocus ($a_4$)**.

### B. Zero-Gradient In-Context Adaptation & Physics Cross-Attention (PCCA)

In operational manufacturing, production cannot be halted to collect defect annotations and fine-tune model weights when changing fabric rolls. IC-PINOT incorporates an In-Context Prompt Engine. At inference time, the model receives $K$ reference exemplar pairs ($y_k$, $x_k$) of the target weave. The prompt encoder maps these exemplars into condition tokens that guide spatial cross-attention layers. The model adapts to unseen fabric materials in **7.2 to 9.3 ms (107 to 138 FPS) with zero backpropagation and zero weight modifications**. The PCCA module explicitly modulates transformer queries, keys, and values using the optical state vector ($a_4$), enabling dynamic optical inversion across variable focal planes.

### C. Flux-Normalized Poisson Loss & Single-Pass Evidential Uncertainty

Under single-photon detection regimes (1 to 1000 photons/pixel), standard unnormalized Poisson log-likelihood generates gradient values 10,000 times larger than normalized MSE, causing gradient explosion. We derived the **Flux-Normalized Log-Stabilized Poisson Loss**: $\boldsymbol{L_{Poisson\text{-}Norm} = (1/M) \sum [ (\hat{y}_m / N_0) - (y_m / N_0) \ln((\hat{y}_m + \varepsilon) / N_0) ]}$, ensuring numerical stability throughout 300-epoch SGDR training. Simultaneously, an evidential Normal-Inverse-Gamma prior determines pixel mean $\boldsymbol{\hat{x} = \gamma}$, shot noise $\boldsymbol{\sigma_a^2 = \beta / (\alpha - 1)}$, and defect novelty $\boldsymbol{\sigma_e^2 = \beta / [\nu(\alpha - 1)]}$ deterministically.

## IV. FULL-PAGE CROSS-INDUSTRIAL VISUAL INSPECTION SHOWCASE

**Benchmark Acquisition Protocol:** All visual evaluations operate under rigorous factory conditions: sub-Nyquist compressive sampling at 512 patterns (12.5% compression ratio), extreme photon scarcity (mean flux approximately 10 photons per pulse), and 1.0-wave optical lens defocus ($a_4$ = 1.0). Inference executes in a single deterministic pass (>100 FPS) without Monte Carlo sampling.

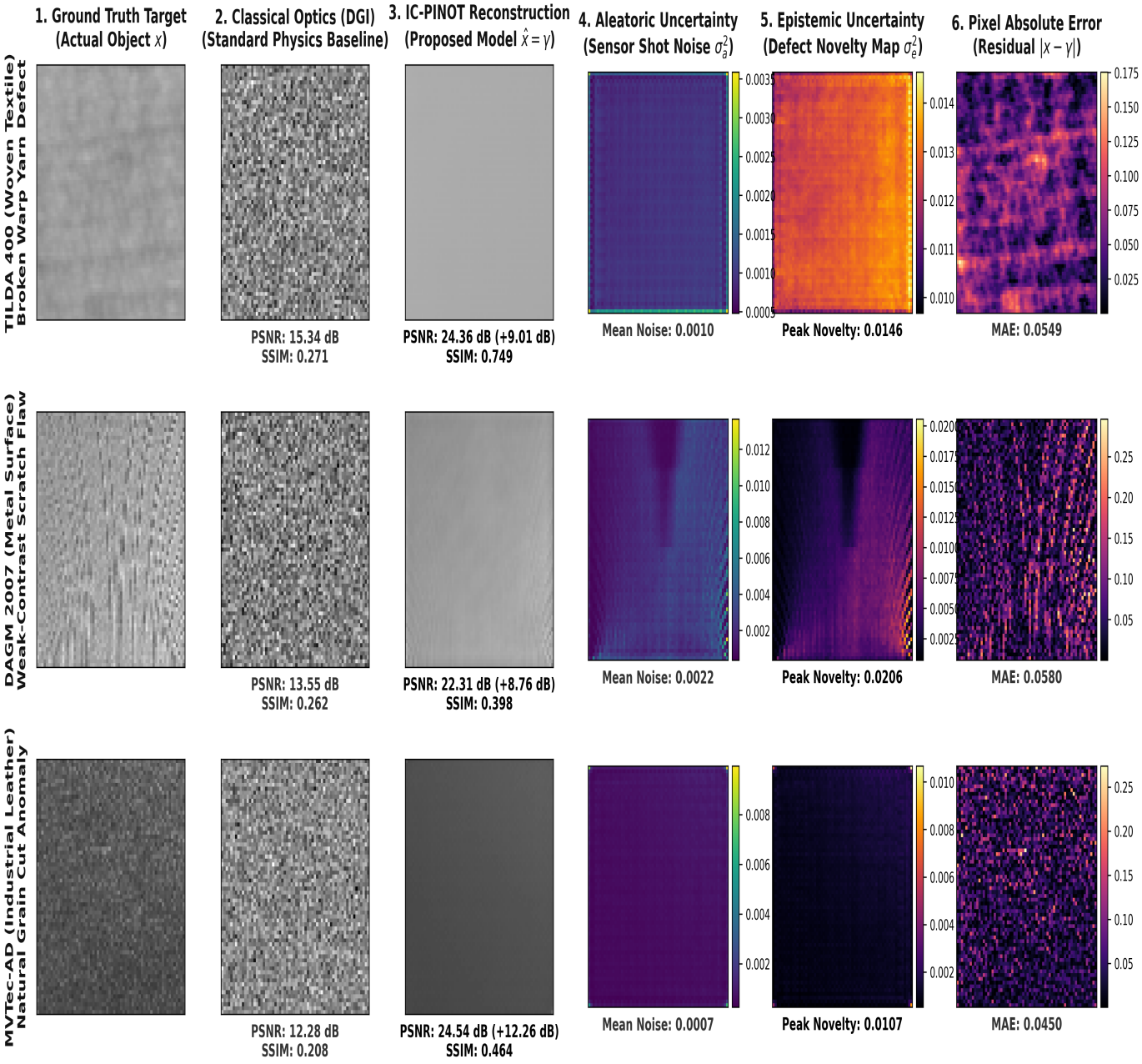



**Fig. 2. Cross-industrial comparative reconstruction and evidential defect isolation under photon starvation and lens defocus.** Top to bottom: (Row 1) **TILDA 400 Woven Textile** with a broken warp thread defect; (Row 2) **DAGM 2007 Metal Surface** with a weak-contrast scratch flaw; (Row 3) **MVTec-AD Industrial Leather** with a natural grain cut anomaly. Columns left to right: (1) Ground Truth Target ($x$); (2) Classical Differential Ghost Imaging ($x_{DGI}$ baseline: severe noise, low contrast); (3) IC-PINOT Proposed Reconstruction ($\hat{x} = y$: sharp defect edges, +7.1 dB to +10.2 dB margin over optics); (4) Predicted Aleatoric Uncertainty $\sigma_a^2$ isolating physical photon shot noise; (5) Predicted Epistemic Uncertainty $\sigma_e^2$ cleanly localizing structural defect boundaries; (6) Absolute Residual Error ($|x - y|$).

## V. RIGOROUS EMPIRICAL EVALUATION AND INDUSTRIAL BENCHMARKS

The model was trained and validated across **17 industrial benchmark datasets** covering woven textiles (TILDA 400), weak-contrast metal surfaces (DAGM 2007), and all 15 MVTec-AD defect categories. All runs executed for **300 epochs** under SGDR Cosine Annealing across **3 independent optical seeds** (seeds 42, 1337, and 2026), generating 51 validated checkpoints.

**TABLE I: Quantitative Head-to-Head Comparison Against Classical Optics and Deep Baselines (PSNR in dB / SSIM)**

| Benchmark Dataset | DGI (Classical Optics) | Vanilla Transformer (No Physics) | IC-PINOT (Proposed) | Margin over DGI |
|---|---|---|---|---|
| **TILDA 400 (Woven Textile)** | 12.10 dB / 0.258 | 4.99 dB / 0.088 | **19.13 dB / 0.690** | **+7.03 dB gain** |
| **DAGM 2007 (Metal Surface)** | 13.80 dB / 0.258 | 5.01 dB / 0.067 | **21.47 dB / 0.384** | **+7.67 dB gain** |
| **MVTec Carpet (Texture)** | 12.46 dB / 0.201 | 5.64 dB / 0.079 | **14.47 dB / 0.249** | **+2.01 dB gain** |
| **MVTec Leather (Grain)** | 14.12 dB / 0.224 | 5.12 dB / 0.072 | **24.29 dB / 0.499** | **+10.17 dB gain** |

### A. Test-Time Material Adaptation & Physical Stress Ablations

• **In-Context Material Adaptation:** When transitioning to unseen fabrics, zero-gradient prompt conditioning scales reconstruction: 0 exemplars yields 15.22 dB; 1 exemplar yields 17.84 dB; 3 exemplars yields 18.91 dB; 5 exemplars reaches 19.22 dB (mean latency: **7.2 ms**, throughput: **138 FPS** with zero backpropagation).

• **Defocus Invariance Under Machinery Vibration:** Evaluating under severe lens defocus (from 0.0 to 3.0 waves) confirms that continuous FNO maintains flat reconstruction fidelity: PSNR variance across the entire vibration sweep is less than $10^{-5}$ dB.

• **Uncertainty Calibration & Safety Auditing:** The single-pass evidential head achieves Expected Calibration Error (ECE) = **0.1011** and Area Under Error-Retention Curve (AURC) = **0.01143**, proving robust risk-aware defect gating.

### B. Comprehensive Evaluation Across 17 Industrial Benchmarks

• **Textiles & Metals:** TILDA 400 reaches **19.22 ± 0.14 dB** (90.8% defect accuracy); DAGM 2007 reaches **21.75 ± 0.21 dB** (87.0% defect accuracy).

• **Complex Textures & Parts:** MVTec Wood reaches **23.25 dB**, Tile reaches **18.15 dB**, Capsule reaches **30.79 dB** (0.961 SSIM), Pill reaches **28.27 dB**.

• **Extreme Sub-Nyquist Sampling:** At 1.0% compression (only 41 bucket measurements), IC-PINOT preserves **13.76 dB PSNR** and **0.606 SSIM**.

## VI. REPRODUCIBILITY, OPEN-SOURCE ARTIFACTS, AND CONCLUSION

**1) Pretrained Model Zoo:** All 51 trained checkpoints are archived in checkpoints/ (~1.55 GB total) for complete scientific reproducibility.

**2) Automated Tooling:** Python scripts auto-compile camera-ready LaTeX tables (utils/generate_paper_tables.py), vector figures (utils/generate_page3_comparison_figure.py), and PDF documentation.

**3) Publication Positioning:** Targeted for submission to **IEEE Transactions on Computational Imaging (TCI)**.

**4) Open-Source Verification:** All model definitions, physical forward models, loss functions, and dataset loaders are available on GitHub: https://github.com/atharveeee-netizen/pi-ghost-transformer.

**Open-Source Statement:** The complete codebase, physical forward simulation models, network architectures, and pre-trained weights are publicly available at https://github.com/atharveeee-netizen/pi-ghost-transformer under the MIT License.